VIT
Vellore Institute of Technology

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BCSE302P - DATABASE SYSTEMS LAB

Semester : Winter Semester 25-26 **Faculty : Dr.S.Geetha**

TASK

1. **While creating any new table suffix your registration number along with it. Eg: EMPLOYEE_24BCE1234**
2. **General Syntax for Table Creation:**
   a. CREATE TABLE TableName (Column1 DataType Constraint, Column2 DataType Constraint, Column3 DataType Constraint, ..., ColumnN DataType Constraint);
   b. TableName: The name of the table you want to create.
   c. Column Definitions: Each column is defined with a name, data type, and optional constraints.
   d. Data Types:
      1. INTEGER: Integer values (e.g., IDs, counts).
      2. VARCHAR(size): Variable-length character strings (e.g., names, emails). Specify the maximum size.
      3. DATE: Dates in YYYY-MM-DD format.

4. DATETIME: Date and time in YYYY-MM-DD HH:MM:SS format.
5. BOOLEAN: True/false values.
6. DECIMAL(precision, scale): Fixed-point numbers for monetary or precise values (e.g., DECIMAL(10, 2) for up to 10 digits, with 2 after the decimal).

STEP 7 — Experiment Questions

- What is Data Manipulation Language (DML)?
- Differentiate between INSERT and UPDATE commands.
- What happens if the WHERE clause is omitted in an UPDATE command?
- How does DELETE differ from TRUNCATE?
- Write the output of the STUDENT table after executing all valid INSERT, UPDATE, and DELETE commands in this experiment.

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Table Structures - *(Suffix your Registration Number to all table names)*

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1. STUDENT

STUDENT (STUD_ID, STUD_NAME, GENDER, DEPT, YEAR, AGE, CGPA, CITY)

Constraints:

- STUD_ID → PRIMARY KEY
- STUD_NAME → NOT NULL
- GENDER → CHECK (GENDER IN ('M','F'))
- YEAR → CHECK (YEAR BETWEEN 1 AND 4)
- AGE → CHECK (AGE > 18)
- CITY → DEFAULT 'CHENNAI'

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2. COURSE

COURSE (COURSE_ID, COURSE_NAME, CATEGORY, DURATION, CREDITS, FEES)

Constraints:

- COURSE_ID → PRIMARY KEY
- COURSE_NAME → NOT NULL
- CATEGORY → CHECK (CATEGORY IN ('UG','PG'))
- DURATION → CHECK (DURATION >= 1)
- CREDITS → CHECK (CREDITS > 0)

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3. FACULTY

FACULTY (FAC_ID, FAC_NAME, GENDER, QUALIFICATION, EXPERIENCE, SALARY, CITY)

Constraints:

- FAC_ID → PRIMARY KEY
- FAC_NAME → NOT NULL
- GENDER → CHECK (GENDER IN ('M','F'))
- EXPERIENCE → CHECK (EXPERIENCE >= 0)
- CITY → DEFAULT 'CHENNAI'

A. INSERT OPERATIONS – 10 TASKS

1. Insert a student record specifying all column values.
2. Insert a student record without specifying CITY (DEFAULT should apply).
3. Insert three student records using a single INSERT statement.
4. Insert a course record using column list in a different order.
5. Insert multiple course records in a single statement.
6. Insert a faculty record specifying all attributes.

7. Insert a faculty record without specifying CITY.
8. Insert a faculty record using substitution variables.
9. Insert records into all three tables ensuring all CHECK constraints are satisfied.
10. Attempt to insert a student with AGE less than or equal to 18 and observe the error.

B. UPDATE OPERATIONS – 10 TASKS

1. Update CGPA of a student using STUD_ID.
2. Update CITY of all students belonging to a specific department.
3. Increment YEAR by 1 for all final-year students.
4. Update COURSE fees by increasing 10% for PG courses.
5. Update COURSE duration for a given COURSE_ID.
6. Update faculty salary based on EXPERIENCE.
7. Update CITY to 'BANGALORE' for female faculty members.
8. Update multiple columns (SALARY and CITY) for a single faculty record.
9. Update student names to uppercase.
10. Attempt to update COURSE category to an invalid value and note the error.

C. DELETE OPERATIONS – 10 TASKS

1. Delete a student record using STUD_ID.
2. Delete all students belonging to YEAR = 1.
3. Delete students whose CGPA is below 5.0.
4. Delete a course using COURSE_ID.
5. Delete all courses with duration less than 2 years.
6. Delete faculty members with EXPERIENCE less than 1 year.
7. Delete faculty belonging to a specific CITY.
8. Delete all student records from the table.

9. Rollback the deletion (if transaction control is enabled).
10. Attempt to delete a record that does not exist and observe the result.

D. SELECT OPERATIONS – 10 TASKS (Single Table Only)

1. Display all student details.
2. Display STUD_NAME, DEPT and CGPA of students.
3. Display students whose AGE is greater than 20.
4. Display students ordered by CGPA in descending order.
5. Display distinct CITY values from STUDENT table.
6. Display all course details.
7. Display courses belonging to PG category.
8. Display faculty details ordered by EXPERIENCE.
9. Display FAC_NAME and SALARY of faculty working in CHENNAI.
10. Display faculty members with salary greater than a given amount.